

Ahmed Iprahim Abdelhamed

Junior Data Scientist

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Portfolio: afadamarcello-code.github.io

SUMMARY

Results-driven Data Scientist with a B.Sc. in Computer Science and ALX certification in Machine Learning and Data Science. Experienced in building end-to-end ML models - including hybrid recommender systems, regression pipelines, and predictive analytics solutions. Strong foundation in Python, Scikit-learn, SQL, and statistical modeling, with hands-on experience applying data-driven approaches to fintech operations at Tabby.

SKILLS

Machine Learning & Modeling: Supervised/Unsupervised Learning, Feature Engineering, Regression/Classification, SVD, Hyperparameter Tuning

Languages & Querying: Python (Pandas, NumPy, Scikit-Learn, Statsmodels), SQL (PostgreSQL, DuckDB), Bash, Git

Visualization & Deployment: Streamlit, Power BI, Looker Studio, Plotly, Matplotlib, Seaborn

Data Engineering & Diagnostics: ETL Pipelines, EDA, VIF Analysis, LINE Framework, Statistical Testing

PROJECTS

CineMatch AI - Hybrid Movie Recommender System | Python, Scikit-learn, Pandas, NumPy, Streamlit

- Built an end-to-end hybrid recommender processing 10M+ MovieLens ratings using SVD (bias + latent factors).
- Achieved an RMSE of 0.8794, outperforming baseline models by 17% through feature engineering and hyperparameter tuning.
- Deployed an interactive Streamlit web application featuring fuzzy string search and real-time similarity matching.

Ames Housing Valuation & Regression Diagnostics | Python, Scikit-learn, Statsmodels, Streamlit

- Built an end-to-end regression pipeline predicting real estate valuations while optimizing model complexity and signal density.
- Reduced feature space from 37 to 24 by diagnosing correlation matrices and Variance Inflation Factors (VIF) to eliminate 13 multicollinear variables.
- Validated LINE regression assumptions and engineered 4 interaction/polynomial features, boosting R^2 from 0.8356 to 0.8999.
- Deployed an interactive Streamlit application enabling real-time valuation estimates for stakeholders.

PROFESSIONAL EXPERIENCE

Tabby | Data & Analytics Specialist | 01/2024 - Present

- Built Python scripts to extract, clean, and aggregate multi-channel social engagement datasets of over 50,000+ records for quantitative performance analysis.
- Automated weekly reporting pipelines using Python and SQL, eliminating manual data collection and reducing reporting time by 10+ hours per week.
- Performed time-series and statistical trend analysis on 10,000+ weekly customer inquiry volumes, accurately forecasting demand spikes and optimizing operational capacity.

Tabby | Social Media & Customer Support Specialist | 01/2023 - 12/2023

- Analyzed historical performance metrics across chat, email, and phone channels to identify service level variance, consistently maintaining 92%+ CSAT.
- Managed real-time interaction datasets and pattern anomalies to maintain SLA adherence during volume spikes.

EDUCATION

Helwan University | Bachelor of Science in Computer Science | 2024 | Helwan, Egypt

CERTIFICATIONS

Data Science Certificate | ALX Africa - 2024

Machine Learning Certificate | ALX Africa - 2024

Data Analytics Professional Certificate | ALX Africa - 2024

Python Programming Certificate | ALX Africa - 2024